

PES UNIVERSITY
Department of Computer Science and Engineering
Software Engineering
UE24CS341A

Software Test Plan (STP)

StreamHub – OTT Video Streaming Website

Project: StreamHub – OTT Video Streaming Website (PES University, UE24CS341A Software Engineering)

Version: 1.0 | **Authors:** TEAM-04 | **Date:** 04-10-2026 | **Status:** Draft for Review

1. Introduction

Purpose & Scope: This plan defines how StreamHub v1.0 — a single responsive website (React frontend, FastAPI backend, PostgreSQL database) — will be tested across its six functional modules: account/authentication & profile, catalog browse & search, playback, personalisation, subscription & payment, and content administration. The payment gateway's internal processing, the email provider's internal delivery, and any native mobile/TV client are excluded, since StreamHub is website-only (SRS §1.2, §2.5).

References: StreamHub SRS v1.0, StreamHub SAD v1.0, OWASP guidelines, PCI-DSS awareness (payment handled entirely by the external gateway).

Definitions: OTT, VOD, JWT, API, PII, RTM, SPA — as defined in the SRS (§1.4).

2. Scope of Testing

Module	Key Requirements Covered
Account, Authentication & Profile	OTT-F-001–004: register + OTP, login, password reset, edit profile
Catalog, Browse & Search	OTT-F-010–012: genre browsing, search, title-detail page
Playback	OTT-F-020–022: video playback, resume position, watch history
Personalisation	OTT-F-030–031: watchlist, 1–5 star rating
Subscription & Payment	OTT-F-040–042: plans, payment-gateway checkout, plan status
Content Administration	OTT-F-050–052: upload title, publish/unpublish, view users

Performance/usability/availability (OTT-NF-001–006) and security (OTT-SR-001–006) requirements are verified within each module above, not as a separate pass.

3. Out of Scope

- Payment gateway's internal payment processing (third-party responsibility; SRS §3.3, §6.2)
- Video transcoding / adaptive-bitrate pipeline (out of scope for this version; SRS §1.2, §2.5)
- Native mobile or smart-TV applications (StreamHub is website-only; SRS §1.2)

- Email service provider's internal delivery infrastructure

4. Test Strategy

Levels:

- Unit tests — module-level (auth, catalog, playback, subscription, admin)
- Integration tests — frontend ↔ backend ↔ database, and backend ↔ payment gateway / email
- System tests — end-to-end website flows
- Acceptance tests — UAT with the course coordinator / project guide

Types: functional, regression, performance, usability, and security testing.

Entry criteria: stable build delivered, test data available, environment ready. Exit criteria: 100% of planned test cases executed, zero critical defects open, RTM fully updated.

Security validation:

- Password handling — bcrypt hashing, no plaintext storage or logging (OTT-SR-002)
- JWT session tokens — short-lived, validated on every protected route (OTT-SR-003)
- HTTPS/TLS 1.2+ enforced on all browser-to-backend traffic (OTT-SR-001)
- Admin-only routes reject non-admin callers (OTT-SR-006)
- API inputs validated via Pydantic models (OTT-SR-005)
- Card/payment data confirmed to never reach or persist on the StreamHub server (OTT-SR-004)

5. Test Environment

Hardware/Browsers: desktop and tablet-width devices; Chrome, Edge, Firefox, Safari.

Software: React SPA, FastAPI backend, PostgreSQL database, payment-gateway sandbox, SMTP test account.

Tools: Playwright/Selenium (UI), Postman (API), JMeter/Locust (performance), GitHub Issues/Jira (defects).

Test Data: dummy viewer accounts, sample catalog titles, sandbox payment cards, test OTP/reset emails.

6. Test Schedule

Milestone	Target Date
Test case design	To be scheduled – before build freeze
Environment setup	To be scheduled – after backend deployment
Test execution (start – end)	To be scheduled
UAT (project guide review)	To be scheduled – before final submission

7. Deliverables & Roles

Deliverables: Test Plan (this document), test cases, test scripts, test data, execution logs, defect reports, final Test Summary Report.

Role	Name	Responsibility
QA Lead	TEAM-04	Prepare plan, coordinate execution
Test Engineer	TEAM-04	Design & execute test cases, log defects
Developer	TEAM-04	Support defect fixes and triage
Project Guide	Arpitha.K Assistant Professor	Approve test results, sign-off readiness

8. Risks & Assumptions

Risk	Mitigation
Payment gateway sandbox instability	Use documented test cards; mock responses for non-payment runs
Tight timeline before submission	Prioritise High-priority requirements first
Test environment downtime (backend/DB)	Maintain a backup environment on a cloud VM

Assumptions: payment-gateway sandbox and SMTP test service remain available throughout testing; PostgreSQL test database is seeded with representative data before execution.

9. Suspension & Resumption

Suspend if: the environment is unavailable for more than 4 hours, or the build is unstable enough to block over 30% of planned test cases. Resume once blocking defects are resolved and the environment is stable.

10. Traceability & Reporting

Requirement	Test Case
OTT-F-001 – Register with OTP	TC-Auth-01
OTT-F-002 – Login	TC-Auth-02
OTT-F-020 – Play Video	TC-Play-01
OTT-F-021 – Resume Playback	TC-Play-02
OTT-F-041 – Payment Gateway Checkout	TC-Sub-02
OTT-NF-001 – Response time $\leq 2s$	TC-Perf-01

Full matrix: see SRS §9. Metrics tracked: % executed, % passed/failed, defect density & aging, requirement coverage — reported periodically and in a final Test Summary Report.

11. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Project Guide	Arpitha.K Assistant Professor	